

PROJECT REPORT: Real-Time Object Detection & Tracking

1. Project Title

Real-Time Object Detection and Tracking using Webcam

2. Objective

The main objective of this project is to detect and track objects in real-time using a webcam. The system processes live video feed and identifies objects using computer vision techniques.

3. Tools & Technologies

- Python
 - OpenCV (cv2)
 - NumPy
 - Pre-trained object detection model (if used)
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4. Project Description

This project captures live video from a webcam and processes each frame to detect objects. The detected objects are highlighted using bounding boxes, and the system continuously tracks them in real-time.

5. Working Methodology

The system works in the following steps:

1. The webcam is activated to capture live video.
2. Each video frame is read and processed individually.

3. The object detection model identifies objects in the frame.
 4. Bounding boxes are drawn around detected objects.
 5. The system continuously updates detection for real-time tracking.
 6. The program can be closed by pressing the 'Q' key.
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6. Features

- Real-time video processing
 - Object detection using computer vision
 - Object tracking with bounding boxes
 - Smooth performance using webcam feed
 - Easy exit control (Q key)
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7. Applications

- Security and surveillance systems
 - Smart monitoring systems
 - AI-based camera applications
 - Traffic and crowd monitoring
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8. Conclusion

This project demonstrates the basic concept of real-time object detection and tracking using computer vision. It provides a foundation for advanced AI applications such as autonomous systems and intelligent surveillance.
